

Pattern Arithmetic: Phase-Colored Geometric Framework for High-Density Information via Unit Probability Spheres

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Abstract

Traditional matrix representations of continuous state spaces suffer from coordinate rigidity and steep spatial scaling costs ($O(N^3)$). This paper introduces *Pattern Arithmetic*, a continuous geometric framework modeling multi-variable information states using vectors on a unit probability sphere $\mathcal{S} \subset \mathbb{R}^3$, augmented by an internal phase-color attribute $\chi \in [0, 2\pi)$. By mapping continuous probability distributions over vector fields and treating color as an independent internal degree of freedom, overlapping states occupy identical spatial coordinates without signal destruction or loss of information density. We formalize fundamental axioms including Information Idempotence ($1 + 1 = 1$), Incremental Subsumption ($A + A.01 = A.01$), and Phase Interference ($A - A = 0$). We demonstrate how this pattern-based algebra offers a parameter-efficient alternative for high-density latent storage, signal unmixing, and physical field modeling.

1 Introduction

1.1 The Problem: Rigid Matrix Grids vs. Continuous Geometric Spaces

For decades, modern computation, physical modeling, and machine learning have relied on matrices as their foundational representation. Linear algebra structures reality by projecting vectors onto fixed, orthogonal coordinate axes (x, y, z) .

While computationally efficient on modern GPU hardware, matrix algebra imposes a rigid, discretized grid onto a physical universe that is fundamentally continuous and dynamic. To capture fine-grained spatial or probabilistic details in a traditional grid, one must increase grid resolution. In three dimensions, doubling spatial resolution increases memory and processing requirements by a factor of $2^3 = 8$.

Furthermore, standard matrix cells store static numerical values. If two independent fields or signals occupy identical spatial coordinates, standard grids force a choice: average the values (losing individual identity), sum them destructively, or expand the matrix into higher tensor dimensions.

1.2 The Phase-Colored Unit Probability Sphere Framework

To overcome these limitations, we propose a shift to continuous geometric state spaces. We model an information system as a **Unit Probability Sphere** ($\mathcal{S}$), bounded within the solid unit ball $B^3 \subset \mathbb{R}^3$. Each state vector $\vec{v}$ carries four continuous properties:

1. **Direction / Orientation:** Encoded by angular coordinates (θ, ϕ) on the sphere.

2. **Magnitude / Likelihood:** Encoded by radial distance $r \in [0, 1]$ from the origin.
3. **Internal Phase / Color (The 4th Dimension):** Encoded by assigning a continuous phase angle $\chi \in [0, 2\pi)$ (or complex phase factor $e^{i\chi}$) to each vector.

In this architecture, “color” acts as an internal degree of freedom embedded directly within the unit ball. Two vectors can occupy identical spatial coordinates (x, y, z) and magnitude r , yet remain completely distinct, non-corrupting states because they possess different phase-color values.

2 Axiomatic Foundations of Pattern Arithmetic

We define six core axioms governing operational behavior within phase-colored unit probability spaces:

2.1 Axiom 1: The Normalized Unit Truth State ($\mathcal{S} = 1$)

Every complete state of information, physical field distribution, or probability density is mapped to a normalized state vector or continuous field on a Unit Probability Sphere $\mathcal{S}$. The total probability measure across the manifold is bounded by unity:

$$\iiint_{B^3} P(x, y, z) dV = 1.0 \quad (1)$$

A single unit probability sphere represents a complete, normalized unit of state truth ($\mathcal{S} = 1$).

2.2 Axiom 2: The Law of Information Idempotence ($1 + 1 = 1$)

Superimposing an exact, duplicate pattern upon itself yields the identical state pattern. Duplicate state reports introduce zero new entropy and add no spatial volume or dimensional measure:

$$\mathcal{S}_A + \mathcal{S}_A = \mathcal{S}_A \implies 1_{\text{truth}} + 1_{\text{truth}} = 1_{\text{truth}} \quad (2)$$

Proof via Vector Normalization: Let Pattern A be represented by a unit state vector $\vec{v}_A = \begin{pmatrix} 0.6 \\ 0.8 \end{pmatrix}$ with Euclidean norm $\|\vec{v}_A\| = \sqrt{0.6^2 + 0.8^2} = 1.0$. Combining two identical state vectors yields:

$$\vec{v}_{\text{combined}} = \vec{v}_A + \vec{v}_A = \begin{pmatrix} 1.2 \\ 1.6 \end{pmatrix} \quad (3)$$

Re-normalizing the state back onto the unit probability manifold restores the baseline state:

$$\vec{v}_{\text{final}} = \frac{\vec{v}_{\text{combined}}}{\|\vec{v}_{\text{combined}}\|} = \frac{1}{\sqrt{1.2^2 + 1.6^2}} \begin{pmatrix} 1.2 \\ 1.6 \end{pmatrix} = \begin{pmatrix} 0.6 \\ 0.8 \end{pmatrix} = \vec{v}_A \quad (4)$$

Thus, duplicate state superposition preserves state identity ($\mathcal{S}_A + \mathcal{S}_A = \mathcal{S}_A$).

2.3 Axiom 3: Incremental State Subsumption ($\mathcal{S}_A + \mathcal{S}_{A.01} = \mathcal{S}_{A.01}$)

When a baseline state $\mathcal{S}_A$ absorbs an updated pattern $\mathcal{S}_{A.01}$ containing all feature vectors of $\mathcal{S}_A$ alongside a phase or structural increment $\delta = 0.01$, the broader state subsumes the redundant baseline:

$$\mathcal{S}_A + \mathcal{S}_{A.01} = \mathcal{S}_{A.01} \quad (5)$$

This reflects non-destructive state evolution, updating information priors without artificially doubling shared background variables.

2.4 Axiom 4: Internal Phase-Color Degrees of Freedom

Every vector within a probability sphere carries spatial orientation (θ, ϕ) , magnitude $r \leq 1$, and an internal continuous phase-color angle $\chi \in [0, 2\pi)$ (or complex factor $e^{i\chi}$):

$$\vec{v} = (r, \theta, \phi, \chi) \quad \equiv \quad \vec{v}_{\text{spatial}} \cdot e^{i\chi} \quad (6)$$

2.5 Axiom 5: Phase-Dependent Vector Interference

The superposition of two state vectors $\vec{v}_1, \vec{v}_2$ with relative phase offset $\Delta\chi = \chi_2 - \chi_1$ is governed by continuous interference:

$$\|\vec{v}_{\text{total}}\| = \sqrt{\|\vec{v}_1\|^2 + \|\vec{v}_2\|^2 + 2\|\vec{v}_1\|\|\vec{v}_2\|\cos(\Delta\chi)} \quad (7)$$

1. **Coherent In-Phase** ($\Delta\chi = 0^\circ$): Constructive reinforcement ($\|\vec{v}_{\text{total}}\| = 2$; normalized state = 1).
2. **Quadrature Shift** ($\Delta\chi = 90^\circ$): Orthogonal fusion ($\|\vec{v}_{\text{total}}\| = \sqrt{2} \approx 1.414$).
3. **Anti-Phase Inversion** ($\Delta\chi = 180^\circ$): Destructive cancellation ($\|\vec{v}_{\text{total}}\| = 0$).

2.6 Axiom 6: Boundary States — The Null (0) and Infinite (∞) Spheres

- **The Null State** ($\mathcal{S} = 0$, **Black**): Formed via exact anti-phase subtraction ($\mathcal{S}_A - \mathcal{S}_A = 0$). Represents an empty sphere where $\|\vec{v}\| = 0$ at all internal coordinates, corresponding to complete destructive cancellation or zero-state equilibrium.
- **The Infinite State** ($\mathcal{S} = \infty$, **White**): Represents maximum entropy and total spectral superposition, where all possible state vectors and phase frequencies co-exist simultaneously across all continuous directions.